Split-track and duplicate treatment, ana578 default, page 1: how it was decided -- every ana573 conclusion re-tested, one mechanism at a time

step	the question, and the test that decided it (ana578)	decision vs ana573 (green = holds, orange = changed)
CLEAN SAMPLE of events	Good TPC segments (bad-segment list: 7% of the segments), ZB crossings [-20, 390] outside NearX0, one vertex per crossing (multi-vertex crossings skipped), then the event cleaning: TFdup (a collision read again in an overlapping trigger frame, TF) and XTF (one track copied into another crossing of the same TF). Test: the like-sign MIXED spike at (dy, dϕ) = (0,0): x7.97 (nothing on) -> x1.19 (TFdup) -> 1.01 ± 0.004 (+ XTF, mixing exclusion). sec 1	TFdup: holds (12.8% of the events; ana573 12.9%) XTF: CHANGED -- 23% of the tracks with >= 2 INTT clusters carry one from another crossing (ana573 0.9%): new in-time INTT rule
LEVEL-0 CUTS events, tracks	Events: |vtx x|, |vtx y| < 0.5 cm, $\|z_{vtx}\|$ < 16 cm (8 cm in Finalize). Tracks: 0.1 < p_T < 20.1 GeV/c, ntpc ≥ 18, |dcaxy|, |dcaz| ≤ 1.5 cm. Test: <N_{trk}> vs crossing per trigger, N-1 per cut: the jet/photon <N_{trk}> bumps near the trigger are tracks with |dcaz| > 1.5 cm. sec 1.7	unchanged; |dcaz| ≤ 1.5 cm alone removes the bumps, the whole crossing range above 2 is usable
AREA A single-track acceptance	A1 SiPhi zones: vertex-ϕ holes with large |dcaxy| (track density and <|dcaxy|> vs vertex ϕ). A2 central-membrane (CM) hole: tracks that cross z = 0 inside the TPC. Test: the bump of pion R_2(y,y) at |y| < 0.2, width scan. In ana578 most of that bump was split tracks, so the CM decision was MOVED after Area B: with all B cleaners a 0.021-0.027 bump is left, and that is the CM hole. A1, A2	A1: CHANGED -- the [94,115) deg hole is gone; only [70,83) masked (2.3% of the tracks; ana573 4.36%) A2: holds -- w in [-0.07, 0.10) (4.32% of the pions; ana573 2.79%); the bump saturates there
AREA B one particle, two tracks	B1 like-sign split band: |dϕ| ~ 0.7° x |1/p_{T1} - 1/p_{T2}| (quantiles); the pregate holds 99.6% of them. B2 π^+ splits more than π^-: 1.80 ± 0.02 (ana573 1.90). B3 SL / SiSplitScore / RadialGap / MVTX match, one path off at a time (bump, survivors). B4 ULS veto: the $\pi^+\pi^-$ C(Q)/(1+r_2) bump at 20-35 MeV, 3.6 -> 1.1-1.3. B5 looper veto: back-to-back opposite-sign pairs vs relative p-sum. Ridge bump: like-sign 0.53 / 0.35 -> 0.026 / 0.028 (LS paths), unlike-sign 0.225 -> 0.030 (ULS veto); each cleaner only its own sign.	B1, B2, B4: hold (pregate, ULS veto SiSplitScore ≥ 0.05) B3: holds -- all 3 paths kept: MVTX path NEEDED, RadialGap nearly redundant (63 extra pairs) B5: holds at 0.08; ~20% of the looper excess lies above the cut (watch)
AREA C two real tracks, lost or merged	C1 tsep (LS two-track merging): S/M vs min |dϕ^*| and dy; LS C(Q): plateau from 20 MeV with the cut. C2 isep (near-vertex loss, LS and ULS): S/M vs |dϕ^*(3 cm)|; the loss sits below 2°. The two-track hole is shallower than in ana573 (0.84-0.87 vs 0.46-0.74 at |dy| < 0.04). C1, C2	C1: holds (|dy| < 0.06, min |dϕ^*| < 2.0°) C2: holds (|dy| < 1.0, |dϕ^*(3 cm)| < 2.0°)
AREA D watch	D1 the ±15° ULS notch: pairs that meet in ϕ between INTT and TPC are lost in the matching. S/M in the 10-20° band vs meeting radius R_m: 0.929 at 0.15-0.20 m (ana573 0.924). No cheap cut. D1	D1: the same as ana573 -- monitored, not cut
V0 RULES	In-peak V0s claim their daughters; $\Lambda/\bar{\Lambda}$ daughters are split partners. Re-test: ana578 vs ana578_nodau (the partner check off), V0-track C(Q) and R_2(dϕ) at low Q / small dy. page 5	re-test running (2026-10-23)

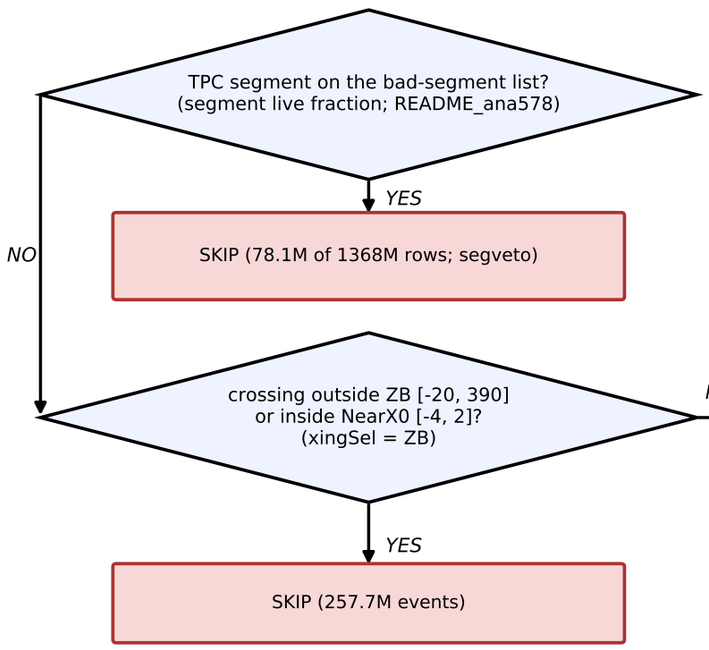
RESULT: the ana573 procedure with the ana573 cut values. Two changes, both forced by the new reconstruction: XTF decides by the in-time INTT cluster when the INTT clusters of a copy disagree, and the SiPhi zone [94,115) is no longer masked. One change of ORDER: the CM mask is decided after Area B. Why it comes out the same: each cleaner targets one MECHANISM of the detector or the reconstruction -- overlapping TFs, a silicon seed shared by two TPC tracks (like- or unlike-sign), loopers, two-track resolution, the near-vertex loss, the CM hole, the silicon-TPC matching gap. ana578 changed the SIZE of several (CM hole half as deep, two-track hole shallower, many out-of-time INTT clusters) but not the mechanisms. Each was re-tested with the symptom it was made for, on the clean output of the step before, so each cut was judged alone -- and the same cuts came back.

Numbers: 6-run production ana578 (136 lists) where given as fractions of events or TF copies; the tests on run 79516 (10M events or all of it, ~280M) in README_SplitTracks578 (section at the end of each box). Pages 2-5: the flowcharts.

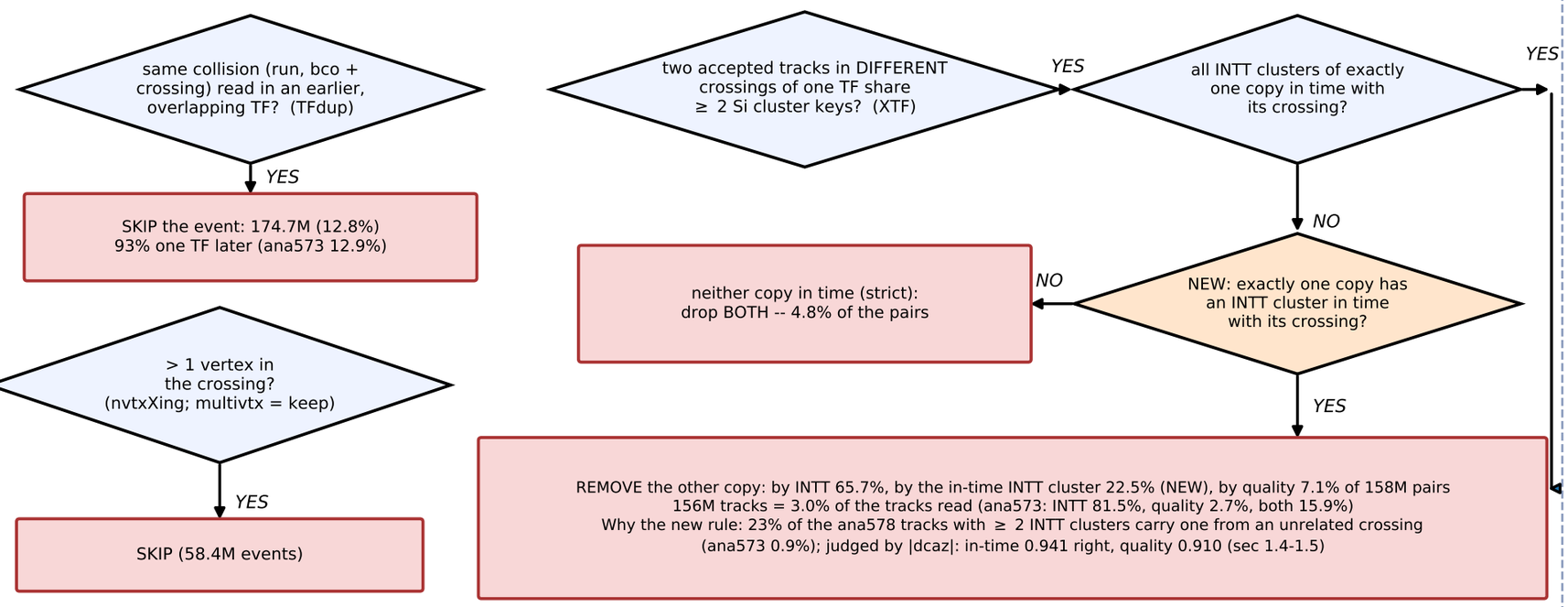
Split-track and duplicate treatment, ana578 default, page 2: the clean sample of events, the level-0 cuts, Area A (single-track acceptance)

CLEAN SAMPLE -- which events can be used at all (event = one crossing with a vertex; the DSTs hold many events per trigger frame, TF)

(a) the event selection (ZB)



(b) event cleaning: pre-passes before the event loop

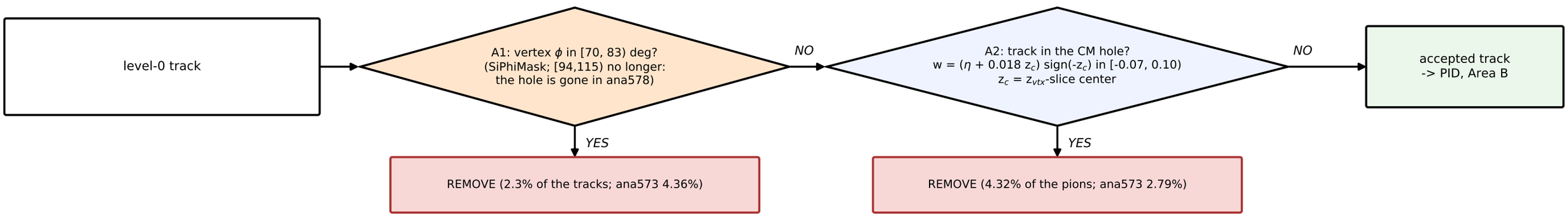


LEVEL-0 CUTS -- per event and per track (AcceptEvent, AcceptTrackBase)

AcceptEvent: |vtx x|, |vtx y| ≤ 0.5 cm, $|z_{vtx}| < 16$ cm in 16 slices (physics: $|z_{vtx}| < 8$ cm, Finalize)
AcceptTrackBase: $0.1 < p_T < 20.1$ GeV/c, ntpc ≥ 18, |dcaxy|, |dcaz| ≤ 1.5 cm
790M events reach the event loop, 490M at $|z_{vtx}| < 8$ cm

Checked on ana578 (sec 1.7, 20M rows of 79516, N-1 per cut and trigger):
the jet/photon $<N_{trk}>$ bumps at |crossing| ~ 30 are vertex-attached tracks with
|dcaz| > 1.5 cm; |dcaz| alone removes them (jet - MB +0.085 -> -0.010 at -20..-5).
86.6% of the Collect tracks pass (ana573 84.0%). Cuts unchanged.

AREA A -- single-track acceptance (AcceptTrack): masks for holes that change from event to event

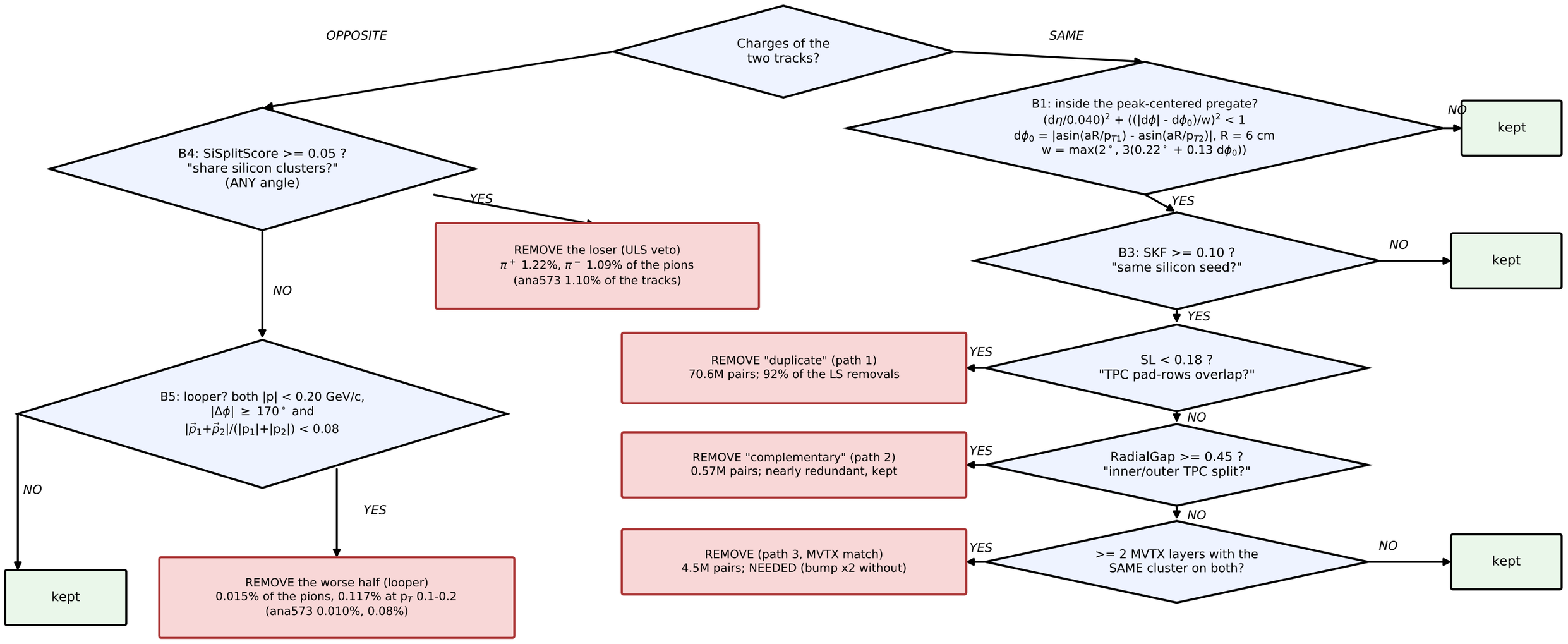


A1 (SiPhi_ana578): [70,83) still a hole with large |dcaxy|, both charges, shallower than in ana573 (2.2-2.3% of the tracks vs 3.6% flat); [94,115) at the flat level, |dcaxy| normal.
A2 (CMMask_ana578, width scan on all of 79516 with every B cleaner): the hole is at the same w, half as deep (minimum 0.42 of the plateau vs 0.23). Without the mask a
charge-independent R_2(y,y) bump of 0.021-0.027 is left after Area B; [-0.07,0.10) takes it to 0.001-0.006 and wider windows gain ~1σ per 1.5% of the pions.
Why after B: in ana578 the y ~ 0 bump with no track cleaner was mostly split tracks (like-sign 0.53 -> 0.03 with the LS paths), so the CM hole could only be judged after them (page 3).

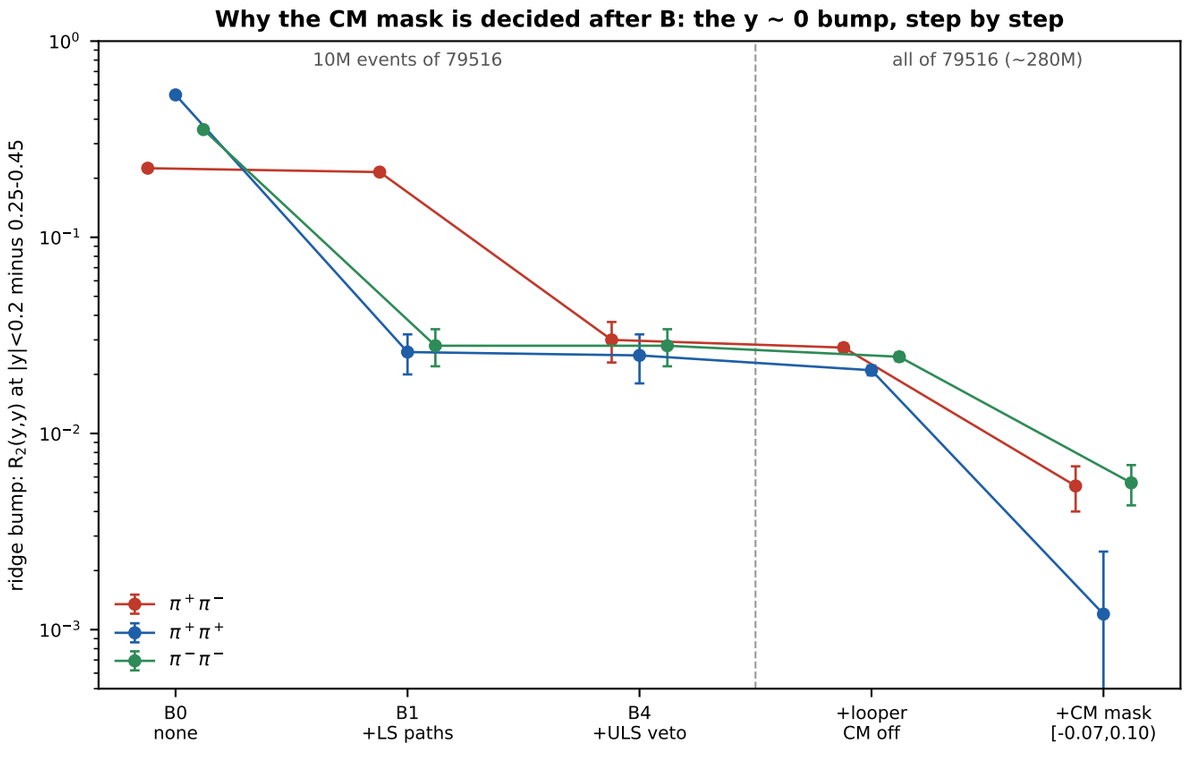
Red = removed / skipped in the ana578 default. Event numbers: 6-run production ana578 (Finalize run record, chunk logs; skips counted in order: TF copies, segments, crossing, multi-vertex). Track fractions: run 79516 studies (README_SplitTracks578 secs 1, A1, A2).

Split-track and duplicate treatment, ana578 default, page 3: Area B -- one particle, two tracks (per event, every pair of accepted charged tracks)

AREA B -- the split-track pre-pass (corral_loop.cxx), before any pair is formed; tested in steps B0 (none) -> B1 (+LS) -> B4 (+ULS veto) -> B5 (+looper)



loser rule: fewer TPC clusters, then worse fit quality (looper: then the larger |dca| sum). LS paths: 1.77% of the accepted pions (6 runs; ana573 1.74% of the tracks).
All of Area B, 6 runs: 202M tracks flagged, 20.2% of the events have at least one flag (ana573 18.3%). A removed track leaves singles, sibling and mixed pairs alike.



Each cleaner removes the bump of its own sign only: the LS paths take $\pi^+\pi^+$ 0.53 -> 0.026 and $\pi^-\pi^-$ 0.35 -> 0.028, the ULS veto takes $\pi^+\pi^-$ 0.215 -> 0.030.
Left after B: a COMMON bump of 0.021-0.027 in all three, the same for like- and unlike-sign -- not a split-track signature. The CM mask takes it to 0.001-0.006: that residual is the CM hole (A2, page 2).

The same B tests, ana578 vs ana573:
B1 split band medians 0.03-0.10 deg higher, same 0.7° x $|1/p_{T1} - 1/p_{T2}|$ law; 99.6-99.7% of the shared-silicon pairs inside the pregate
B2 π^+/π^- splits 1.80 ± 0.02 (ana573 1.90)
B3 no central excess without shared silicon; 764 in-gate survivors share exactly one MVTX cluster and leave no visible symptom
B4 $\pi^+\pi^-$ C(Q)/(1+r_2) at 20-35 MeV 3.6 -> 1.1-1.3; like-sign unchanged
B5 74.6k looper pairs over a control of 28.7k below 0.08; ~20% above (watch)

Fractions: 6-run production ana578 where marked; else run 79516 (B4 10M, B5 all of 79516; README_SplitTracks578 secs B0-B5). Pair counts are pairs flagged in the 6-run production (chunk logs).

Split-track and duplicate treatment, ana578 default, page 4: Area C (two real tracks), the crossing correction, Area D and the watch list

AREA C -- pair level (CalcRm::PairInfo): two real tracks, lost or merged; sibling AND mixed pairs alike, in this order; then the maps

C1: LS charged pair with |dy| < 0.06 and
min over R = 0.30-0.78 m of $|d\phi^*|$ < 2.0° ?
(two-track resolution, tsep)

NO

C2: charged pair with $|d\phi^*(R = 3\ cm)|$ < 2.0°
at |dy| < 1.0, LS or ULS ?
(near-vertex two-track cut, isep)

NO

Mixed EVENT pair from
adjacent trigger frames?
(mixing exclusion, with XTF)

YES

REJECT the pair
LS pions: 0.36-0.38% of sibling / 0.36% of mixed pairs
(ana573 0.35% / 0.33%)

YES

REJECT the pair
ULS pions 0.82% S / 0.89% M; LS 0.84% / 0.88%
(ana573 0.86 / 0.94; 0.89 / 0.93)

YES

SKIP the event pair
same-TF share of the mixed
pairs left: 5.1% (ana573 5.8%)

then the (dy, dϕ) maps get the CROSSING CORRECTION (pt ordering; the dirty dϕ side <- its clean mirror). A correction, not a removal:
it adds ~0.2% of the unlike-sign and 0.1-0.15% of the like-sign pairs, the same as in ana573, flat in event rate (CrossCompare_ana578.pdf).

AREA D and the WATCH LIST -- monitored in every production, not removed

D1: the ±15° ULS notch (ST578_D1_Notch.pdf)
ULS S/M in the 10-20° band / 25-50°: 0.971 (ana573 0.955-0.975).
vs the radius where the two tracks meet in ϕ, R_m:
0.970, 0.941, 0.929 (0.15-0.20 m), 0.932, 0.950, 0.981 (0.30-0.35)
= ana573 (0.974 0.936 0.924 0.929 0.944 0.983): unchanged by the
ana578 reconstruction. The loss is in pairs that MEET inside the gap
(min $|d\phi^*|$ < 0.5° over 0.10-0.30 m: 0.937) and does not shrink with
dy (0.952 at |dy| < 0.1, 0.919 at 0.9-1.0): the matching seems to use
ϕ only -- a note for tracking. LS has a shallow notch too (~0.98-0.99).
A meeting-radius cut would drop 12.9% of the near ULS pairs and leave
a ~2% notch: no cut.

Watch items (recheck with all 6 runs)
- $\pi^-\pi^-$ R_2 on the $y_1 = y_2$ diagonal +0.003 above $\pi^+\pi^+$ everywhere,
+0.0035 ± 0.0018 more at |y| < 0.2, with and without the CM mask (A2)
- the two-track dip at |dy| 0.10-0.14, < 1°: S/M 0.92-0.94, beyond any
tsep window, and the 0.07 gap left in the central dy bin after tsep (C1)
- the looper excess above the 0.08 cut, ~20% (B5)
- the crossing correction per production: mixed-species unlike-sign pairs
flip between ana573 and ana578 (π^+K^- vs π^-K^+, 3-5σ)
- V0 masses and C(Q) peak positions (momentum scale)
- tracks removed per cleaner: the chunk logs and Finalize

Red = removed / skipped in the ana578 default. Pair-cut fractions: 6-run production ana578 (pair-cut counters of the 136 chunk logs, pions). Notch and watch items: all of 79516 (README_SplitTracks578 secs C1, C2, D1, A2, B5; crossing step).

Split-track and duplicate treatment, ana578 default, page 5: V0s -- which V0s pair, and what happens to their daughters

STAGE V1 -- per event, every KFP V0 candidate (K^0_S, Λ, $\bar{\Lambda}$)

V0 candidate
(zero-sentinel eta/phi/mass/ctau
rebuilt, README_v0etaSpike)

within $\pm 3\sigma$ of its mass peak?
K^0_S 0.4884 ± 3(0.0076), Λ 1.1143 ± 3(0.0035)
$\bar{\Lambda}$ 1.1134 ± 3(0.0028) GeV
(V0PEAK_*, fluct_common.h)

YES

IN PEAK: enters the V0 pair types and
CLAIMS its two daughters
K^0_S 84.6%, Λ 40.8%, $\bar{\Lambda}$ 29.3% of the candidates
(ana573 84.5, 39.9, 28.8%)

NO

OFF PEAK: not paired (it would pair with its own
daughter); its daughters stay ORDINARY tracks

STAGE V2 -- per event, the in-peak daughters in the track machinery (corral_loop.cxx)

in-peak daughter: OUT of every track pair type,
OUT of the split-track candidate list,
so NEVER a split-track loser

parent V0?

Λ, $\bar{\Lambda}$

Λ, $\bar{\Lambda}$: another accepted track
is a SPLIT COPY of a daughter?
same charge: LS paths and $\Delta R < 0.05$
opposite charge: SiSplitScore $\geq$ 0.05;
a proton-tagged partner of the (anti)proton
daughter is its wrong-sign copy

YES

REMOVE the other track
(the daughter wins)
6 runs: LS 267,307,
ULS 222,477 partner pairs

K^0_S

K^0_S: NO partner check
(ana573: it cut real $\pi^\pm K^0_S$ pairs)

Rules from ana573 (README_CQcomparison sec 2.2; v9 -> v13): the partner check removed the pΛ and $\pi\Lambda$ excesses at Q < 0.2 GeV;
checking K^0_S daughters too cut ~10% of the real $\pi^\pm K^0_S$ sibling pairs at Q < 0.4 GeV.

RE-TEST ON ana578 (2026-10-23) -- ana578 (default) vs ana578_nodau (partner check off), same 136 lists

Test: V0-track C(Q)/(1+r_2) at Q < 0.4 GeV and $R_2(d\phi)$ at small dy, for pΛ, $\bar{p}\bar{\Lambda}$ (own-copy possible), p$\bar{\Lambda}$, $\bar{p}\Lambda$ (controls),
$\pi^\pm\Lambda$, $\pi^\pm\bar{\Lambda}$, $\pi^\pm K^0_S$ (K^0_S has no partner check: must not move).
Result: PENDING (array 40752248, Finalize after it).

Green = kept / paired, red = removed, yellow = watched or pending. V0 numbers: 6-run production ana578 (chunk logs). Mass windows unchanged from ana573.